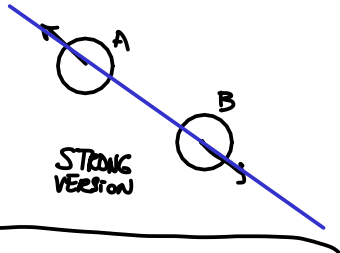
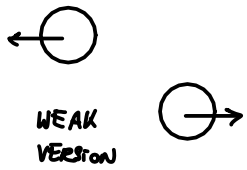
Newton's Laws: 1st Law IF $\sum_i \vec{F}_i = \vec{0} \Rightarrow \vec{v} = \text{CONSTANT}$

2nd Law $m\vec{a} = \sum_i \vec{F}_i$

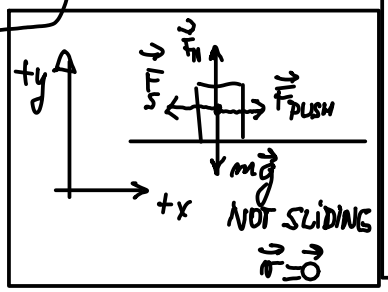
3rd Law

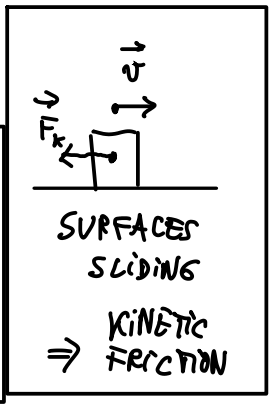




Friction

- STATIC $|\vec{F}_s| \leq \mu_s N = \mu_s F_N$ — TAKES THE VALUE NEEDED SO THAT SURFACES DON'T SLIDE
- KINETIC $|\vec{F}_k| = \mu_k N = \mu_k F_N$ $\quad 0 \leq \mu_k \leq \mu_s$
- ~~ROLLING~~ (IGNORE IN THIS CLASS)
- DRAG $|\vec{F}| \propto f(|\vec{v}|)$





Newton's 2nd Law:

$0 = ma_x = F_{PUSH} - F_s$ (I) $\Rightarrow$ $F_s = F_{PUSH} \leq \mu_s F_N$

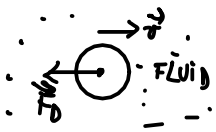
$0 = ma_y = F_N - mg$ (II) $\Rightarrow$ $F_N = mg$

$\Rightarrow F_s \leq \mu_s mg$

WE COULD WRITE, INSTEAD OF (I): $ma_x = F_{PUSH\,x} + F_{Sx}$ (I') BUT IT IS LESS USEFUL BECAUSE IT DOESN'T TELL US THE SIGNS OF F_{PUSHx} AND F_{Sx}

FOR ANY VECTOR: COMPONENTS CAN BE $\gtrless 0$ OR < 0; MAGNITUDES ARE ≥ 0

Drag Force:

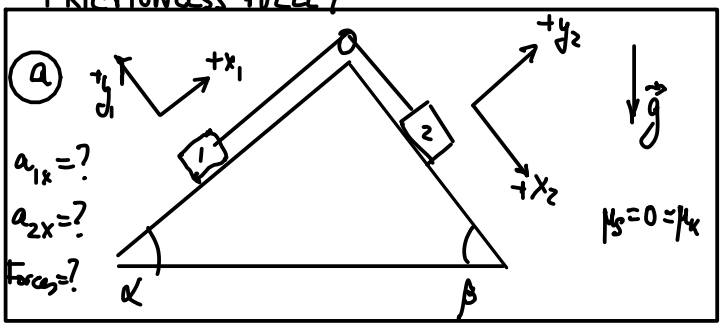


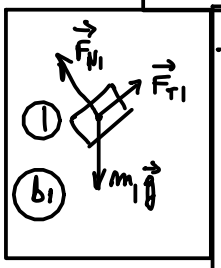
SLOW: $\vec{F}_D = -C_1 \vec{v}$ (BIOLOGY)

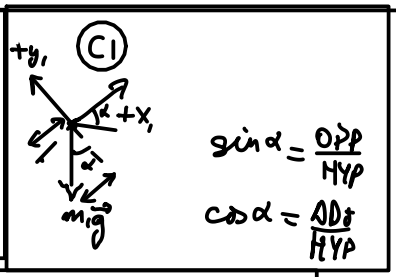
FAST: $F_D \propto K v^2$ (AIRPLANE, CAR)

EXAMPLE 1:

MASSLESS STRING AND PULLEY, STRING INEXTENSIBLE, FRICTIONLESS PULLEY

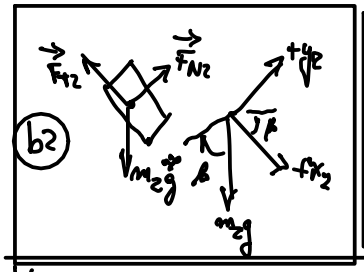






$m_1 a_{1x} = F_T - m_1 g \sin\alpha$

$0 = m_1 a_{1y} = F_{N1} - m_1 g \cos\alpha$ (#1)



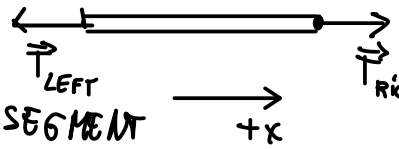
$m_2 a_{2x} = -F_T + m_2 g \sin\beta$

$0 = m_2 a_{2y} = F_{N2} - m_2 g \cos\beta$ (#2)

F_T same for #1 and #2 because string is massless, and pulley massless and frictionless

AUG 26 2026

ASIDE: MASSLESS STRING $\Rightarrow$ TENSION SAME ANYWHERE ALONG STRING



$0 = ma_x = T_{RIGHT} - T_{LEFT} \Rightarrow T_{RIGHT} = T_{LEFT}$ ($m = 0$)

$dx_{RIGHT} = dx_{LEFT} \Rightarrow \left.\frac{dx}{dt}\right|_R = \left.\frac{dx}{dt}\right|_L$ ($v_R = v_L$) $\xrightarrow{d/dt}$ $a_R = a_L$

$dx_{RIGHT} = dx_{LEFT}$ AT ALL TIMES; $\left.\frac{dx}{dt}\right|_R = \left.\frac{dx}{dt}\right|_L$ AT ALL TIMES; $a_R = a_L$ AT ALL TIMES